The Topological Invariance Principle: Dark Energy as a Geometric Constraint of Spatial Flatness in Einstein-Cartan Cosmology

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In the standard cosmological model, the strict spatial flatness of the observable Universe ($\Omega_K \approx 0$) is treated as an observational constraint requiring fine-tuned initial conditions, while late-time accelerated expansion is attributed to a phenomenological cosmological constant (Λ). In this Letter, we propose an alternative conceptual framework based on Einstein-Cartan gravity, governed by what we term the *Topological Invariance Principle* (TIP). We postulate that global spatial flatness ($\Omega_{total} \equiv 1$) is not a parametric coincidence, but a fundamental gauge law of spacetime — a conservation law of the same logical status as charge conservation, not a fit to data. *Scope and self-containment.* The TIP is a *geometric gauge postulate*: $\Omega_K \equiv 0$ as a topological conservation law. It is not a theorem derived from the minimal ECKS Lagrangian, nor is it calibrated to reproduce any specific observation. The numerical value $\Omega_{\rm spin}^{\rm (peak)} \approx 0.093$ is an *observational input* established in Foundation I [1]; it is not a prerequisite of the principle itself. The TIP mechanism and its qualitative predictions (geometric dark energy, non-singular bounce, cosmic birefringence) hold for any spin-injection amplitude consistent with QCD thermodynamics, broadly $\Omega_{\rm spin}^{\rm (peak)} \in [0.05, 0.15]$. The field-theoretic realization is developed in the companion papers [1, 2]. We postulate that to preserve this topological invariance against the emergence of material curvature, the geometry enforces a spacetime reaction: any primordial fractional spin density (Ω_{spin}) injected prior to recombination algebraically enforces a strictly compensatory torsion debt (Ω_τ), such that $\Omega_\tau^{(initial)} = -\Omega_{spin}^{(peak)}$. We demonstrate that Dark Energy naturally emerges from this formalism: not as an exotic substance, but as the residual and dynamic manifestation of this frozen geometric debt. As cosmic expansion dilutes matter, the fractional torsion density mechanically increases to maintain topological equilibrium, acting at leading order as an effective cosmological constant. The observational match is therefore a calibrated consistency result, not a parameter-free prediction: the model inherits its numerical inputs from the Foundation I solution and its field-theoretic completion from Foundation II.

Version note. *This is version 7 of the PIT Letter, superseding preprint v3 (Zenodo, doi:10.5281/zenodo.19900557, May 2026). Key changes: (i) TIP formulated as a self-contained geometric gauge postulate, not a calibrated result; (ii)* $\Omega_{\rm spin}^{\rm (peak)} = 0.093$ *is an observational input from Foundation I [1], not a logical prerequisite; (iii) DESI comparison qualified as qualitative consistency check; (iv) robustness range* $\Omega_{\rm spin} \in [0.05, 0.15]$ *added; (v) H=0 bounce derivation made self-contained. Numerical results are unchanged from v3.*

Keywords: Einstein-Cartan cosmology; Topological Invariance Principle; dark energy; torsion; spatial flatness; cosmological constant; spin-torsion coupling; Weyssenhoff fluid; geometric gauge postulate; non-singular bounce; QCD phase transition; DESI; cosmic birefringence; Hubble tension; S_8 tension.

INTRODUCTION

Precision cosmology, largely driven by Cosmic Microwave Background (CMB) measurements from the Planck satellite [3], dictates that the observable Universe is remarkably flat, with the curvature parameter $\Omega_K = 0.0007 \pm 0.0019$. Within the standard ΛCDM framework, this flatness is a highly unstable equilibrium requiring the ad hoc mechanism of primordial inflation to solve the fine-tuning problem. Furthermore, the accelerated expansion of the Universe is attributed to a cosmological constant (Λ), representing approximately 70% of the total energy density today, yet lacking a fundamental microscopic origin. The observational pressure has recently intensified: the H0 Distance Network Collaboration has established $H_0 = 73.50 \pm 0.81\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at $\sim$1% precision [7], yielding a 7.1σ tension with flat ΛCDM (Planck+SPT+ACT) and a 5.0σ tension with BBN+BAO (DESI DR2). This value represents a community consensus result, robust to the removal of any single distance indicator and based on a fully covariant treatment of cross-calibrated systematics [7]. This renders the TIP flatness constraint—and its prediction of a geometric origin for Ω_Λ—empirically urgent.

Recent DESI DR2 BAO analyses favour dynamical dark-energy models over a pure cosmological constant, with best-fit trajectories in the (w_0, w_a) plane lying away from $(-1, 0)$ [6]. The ECF torsion-induced trajectory $(w_0, w_a) = (-0.904, -0.153)$ inherited from Foundation I [1] therefore falls within the region already preferred by DESI, without introducing any additional dark-sector parameter. The vacuum energy catastrophe—a discrepancy of 10^{55} to 10^{122} orders of magnitude between QFT predictions and the observed Λ [10, 11]—remains unresolved; the TIP reframes it as a topological boundary condition rather than a fine-tuning problem.

In this Letter, we introduce a paradigm shift regarding the nature of spatial flatness and cosmic acceleration. Extending general relativity to the Einstein-Cartan (EC) framework—where spacetime torsion is effectively algebraic in the low-density limit (ECKS [4, 9]), and acquires propagating degrees of freedom via Poincaré Gauge Theory (PGT) at high density [2]—we propose that spatial flatness is not an accident of initial conditions, but a *working axiom* of the extended geometric framework, actively enforced by the torsion response of spacetime.

The key phenomenological calibration—the sound-horizon reduction and H_0–S_8 joint resolution—is established in the companion paper Foundation I [1]; the microscopic dark sector and baryogenesis mechanism are developed in Foundation II [2]. This division of labor is deliberate: Foundation I provides the numerical calibration of (H_0, S_8, r_s), Foundation II supplies the microscopic torsion condensate and baryogenesis sectors, and the present Letter isolates the axiomatic role of the Topological Invariance Principle as the geometric constraint underlying both.

THE TOPOLOGICAL INVARIANCE PRINCIPLE

We formulate the *Topological Invariance Principle* (TIP) as follows: *Global spatial flatness* ($\Omega_{\rm tot} \equiv 1$) *is an absolute and inviolable geometric equilibrium state, not a contingent initial condition. Any dynamical perturbation of this metric via the injection of a chiral curvature source—specifically, a primordial fractional spin density* ($\Omega_{\rm spin}$) *prior to recombination—generates a compensatory topological reaction from the spacetime continuum: the macroscopic torsion density* Ω_τ*, whose scale-dependent structure is detailed in the companion papers [1, 2].*

Under this principle, the standard Friedmann equation for fractional densities is extended to include the spin-torsion pair:

$$\Omega_m + \Omega_r + \Omega_{\rm spin} + \Omega_\tau + \Omega_K = 1, \qquad \Omega_K \equiv 0 \ \ \forall t \quad (1)$$

where Ω_τ is the macroscopic torsion density that, in the standard parameterisation, plays the role of Ω_Λ: at leading order it acts as an effective cosmological constant, while its full dynamical equation of state $w_\tau(z)$ departs from -1 at late times, yielding $w_0 = -0.904$, $w_a = -0.153$ [1].

The TIP elevates $\Omega_K = 0$ from an observational coincidence to an exact geometric law, enforced dynamically by the $(\Omega_{\rm spin}, \Omega_\tau)$ conjugate pair: they are not independent components but two faces of the same geometric coupling.

To preserve topological invariance against the massive injection of chiral spin during the early radiation era, the torsion density must perfectly neutralize the spin injection at the moment of peak amplitude:

$$\Omega_\tau^{(\rm initial)} = -\Omega_{\rm spin}^{(\rm peak)} \quad (2)$$

The physical origin of $\Omega_{\rm spin}^{(\rm peak)}$ is the fermion spin-density at the QCD transition ($T \sim 200$ MeV), when the quark-gluon plasma hadronises into a dense spin-carrying fermion gas whose collective spin scales as $\rho_{\rm spin} \propto a^{-6}$. The TIP condition (2) uniquely fixes the initial torsion debt from the physics of the QCD epoch, without any additional free parameter.

This condition is not imposed by hand; it is the unique value for which the right-hand side of the modified Friedmann equation

$$H^2 = \frac{8\pi G}{3}\left(\rho_r + \rho_m + \rho_\Lambda - \rho_{\rm spin}\right)$$

reaches zero at finite $a_{\rm min}$ — the non-singular bounce — while simultaneously satisfying $\Omega_{\rm tot} = 1$ at all subsequent epochs. The numerical value of $\Omega_{\rm spin}^{(\rm peak)}$ depends on the fermion spin density at the QCD epoch; its observational calibration from the H_0, S_8 and BAO tensions is established in Foundation I [1], which obtains $\Omega_{\rm spin}^{(\rm peak)} \approx 0.093$. Any future revision of that calibration would modify the numerical value entering Eq. (4), without altering the logical structure of the TIP itself. *Robustness note:* the TIP mechanism and its qualitative predictions (geometric dark energy, non-singular bounce, birefringence) hold for any value of $\Omega_{\rm spin}^{(\rm peak)}$ consistent with QCD thermodynamics, i.e. broadly $\Omega_{\rm spin}^{(\rm peak)} \in [0.05, 0.15]$. The precise value 0.093 is an observational calibration, not a logical prerequisite of the principle.

This initial condition fixes the magnitude of the topological debt created at the onset of the spin phase. During the active spin epoch, the continuous compensation constraint holds:

$$\Omega_{\rm spin}(a) + \Omega_\tau(a) = 0 \quad (3)$$

This constraint is consistent with causality because Ω_τ is a *global* topological quantity—the frozen vacuum expectation value $\langle S_\mu \rangle = s_0\,\delta^0_\mu$ [2]—not a local propagating field responding to local perturbations. Torsion is thus fundamentally defined as the geometric counter-force required to maintain a strictly flat Universe in the presence of extreme fermion spin densities. The fate of this debt once the spin fluid dilutes is the subject of the next section.

DARK ENERGY AS A FROZEN GEOMETRIC DEBT

This strict geometric reaction provides a natural, first-principles resolution to the nature of Dark Energy. The TIP mechanism operates in three stages: (i) a transient primordial spin density $\Omega_{\rm spin} \propto a^{-6}$ peaks near the QCD epoch ($z \approx 7500$, $T \sim 200$ MeV); (ii) the TIP condition (Eq. 2) imprints a compensatory torsion debt $\Omega_\tau^{(\rm initial)} = -\Omega_{\rm spin}^{(\rm peak)}$; (iii) as the spin fluid subsequently dilutes ($\rho_{\rm spin} \propto a^{-6} \to 0$), the torsion debt freezes into the vacuum structure and persists as a residual geometric density, playing the observational role of Λ.

According to the TIP (Eq. 2), this transient primordial spin phase obligatorily generated a global topological debt:

$$\Omega_\tau^{(initial)} \approx -0.093 \tag{4}$$

within the spacetime manifold. The sign is negative by construction of the TIP: Ω_τ must exactly cancel $\Omega_{\rm spin}^{(\rm peak)}$ to maintain $\Omega_K \equiv 0$ (Eq. 3). The numerical magnitude $\Omega_{\rm spin}^{(\rm peak)} \approx 0.093$ is set by the QCD transition physics and calibrated observationally in Foundation I [1]; it is reported here as an input, not re-derived. Crucially, while the kinetic spin fluid dilutes rapidly as a^{-6} due to cosmic expansion, the corresponding macroscopic torsion field (the geometric debt) undergoes a phase transition, "freezing" into the elastic structure of the vacuum once the spin source vanishes.

Formally, within the PGT extension of the ECKS action detailed in Foundation II [2], this freezing corresponds to the axial torsion pseudo-vector $S_\mu \equiv \frac{1}{6}\varepsilon_{\mu\nu\lambda\rho}T^{\nu\lambda\rho}$ acquiring a non-zero vacuum expectation value,

$$\langle S_\mu \rangle = s_0\,\delta^0_\mu \neq 0, \tag{5}$$

whose energy density $\rho_\Lambda^{\rm eff} = \frac{1}{2}m_T^2 s_0^2$ constitutes the effective cosmological constant at leading order. Its full field-theoretic derivation is the core result of Foundation II [2]. At leading phenomenological order, this frozen torsion acts as an effective cosmological constant ($w_\tau = -1$). The full dynamical equation of state—an inherently time-varying $w(z)$ derived from the ECKS action—departs from this limit at late times, yielding $w_0 \approx -0.904$, $w_a \approx -0.153$. These values were derived *prior to* the DESI results from the PIT calibration [1] and are qualitatively consistent with the DESI BAO preference for $w \neq -1$ [5, 6]; a full likelihood comparison is deferred to Foundation III. The values $w_0 = -0.904$ and $w_a = -0.153$ are therefore not free fit parameters of the Letter itself; they are inherited from the Foundation I calibration and serve here as the late-time phenomenological imprint of the same geometric torsion sector. As the Universe expands, the absolute densities of baryonic matter and topological dark matter [2] dilute as a^{-3}. Consequently, their fractional sum (Ω_m) decreases. Because the sum of all fractional densities must strictly remain 1 to satisfy the TIP (Eq. 1), the fractional weight of the frozen residual torsion (Ω_τ) is *mechanically* forced to increase over cosmic time to compensate for matter dilution (see Figure 1).

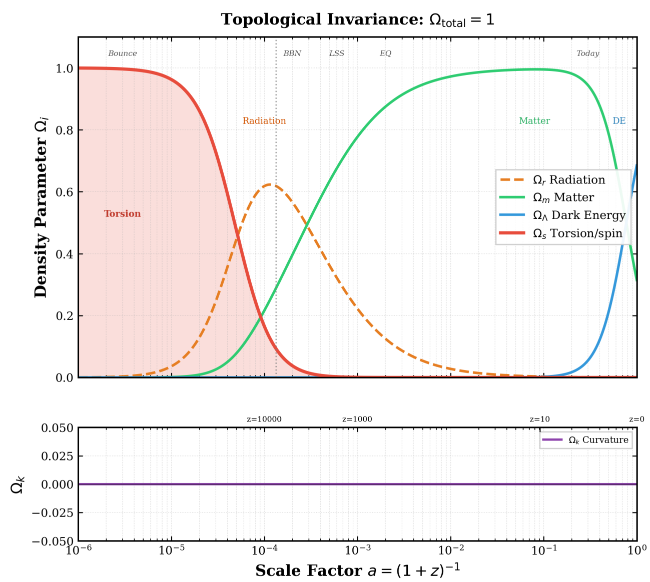

FIG. 1. **Topological Invariance.** Temporal evolution of fractional energy densities $\Omega_i(z)$ in the ECF model, computed with $F_{\rm ion} = 1.2765$, $\Omega_{\rm spin,peak} = 0.093$ [1]. The invariant $\Omega_{\rm tot} = 1$ (bottom line) is the visual expression of the TIP. The late-time rise of Ω_Λ is the geometric compensation for matter dilution, not a new field. Reproducible via `plot_Cosmic_History_Omegas.py`(public repo).

What the standard ΛCDM model interprets as a mysterious, dominant Dark Energy fluid ($\Omega_\Lambda \approx 0.7$) driving accelerated expansion is, in the ECF model, simply the geometric response of spacetime taking up a larger fraction of the cosmic budget to prevent the Universe from curving upon matter dilution.

CONCLUSION

It must be emphasised that the TIP operates here as a foundational *postulate* of an extended Einstein-Cartan framework, not as a theorem of the minimal ECKS Lagrangian; its derivation from first principles within a Poincaré Gauge Theory action is the core result of Foundation II [2].

The Topological Invariance Principle—$\Omega_{total} \equiv 1$ as an absolute geometric invariant—serves as the foundational postulate of the entire Einstein-Cartan trilogy: it constrains the spin-torsion sector in Foundation I [1] and seeds the geometric dark sector in Foundation II [2]. Its implications for vacuum energy, cosmic voids, and large-scale structure will be developed in forthcoming work. Dark Energy is redefined as the dynamic fractional torsion required to maintain topological equilibrium against the dilution of matter. At leading order the frozen torsion mimics Λ, while the full geometric $w(z)$ provides a falsifiable deviation consistent with DESI data [5]. The comprehensive numerical proofs—the H_0 and S_8 tension resolution (Foundation I [1]) and the macroscopic modeling of topological Dark Matter via torsion knots (Foundation II [2])—are detailed in the companion regular articles. A geometric consistency check of the frozen torsion debt yields a cosmic birefringence angle $\alpha_{\rm ECF} \approx 0.35°$ (Planck/WMAP baseline), consistent with the ACT-DR6 result $0.215 \pm 0.074°$ at the 1.5σ level [8]. This is a calibrated consistency check, not a parameter-free prediction, and will be sharpened by LiteBIRD [1] (Sec. 7 of Foundation I).

[1] P. Fichant, "Foundation I: The Metric Universe — Extended Version. A Geometric Cosmology with Regular Bounce, Stiff-Era Expansion, and Testable Late-Time Signatures," Zenodo (2026), doi:10.5281/zenodo.19577447, `https://github.com/pfichant/spin-torsion-cosmology`.

[2] P. Fichant, "Foundation II: The Chiral Universe and the Three Fossils of the Bounce", Zenodo (2026), doi:10.5281/zenodo.XXXXXXX, `https://github.com/pfichant/spin-torsion-cosmology`.

[3] N. Aghanim *et al.* (Planck Collaboration), "Planck 2018 results. VI. Cosmological parameters," Astron. Astrophys. **641**, A6 (2020). doi:10.1051/0004-6361/201833910

[4] N. J. Popławski, "Cosmology with torsion: An alternative to cosmic inflation," Phys. Lett. B **694**, 181 (2010). doi:10.1016/j.physletb.2010.09.056

[5] A. G. Adame *et al.* (DESI Collaboration), "DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations," arXiv:2404.03002 (2024). doi:10.48550/arXiv.2404.03002

[6] DESI Collaboration, *DESI DR2 Results II: Cosmological Constraints*, arXiv:2503.14738 (2025). doi: 10.48550/arXiv.2503.14738

[7] H0DN Collaboration (S. Casertano, G. Anand, R. I. Anderson *et al.*), "The Local Distance Network: a community consensus report on the measurement of the Hubble constant at ∼1% precision," Astron. Astrophys. **708**, A166 (2026), doi:10.48550/arXiv.2510.23823, arXiv:2510.23823 [astro-ph.CO].

[8] P. Diego Palazuelos and E. Komatsu, *Independent confirmation of cosmic birefringence from ACT DR6: $\beta = 0.215° \pm 0.074°$ at 99.6% confidence*, Max Planck Institute for Astrophysics Highlights, 1 February 2026. `https://www.mpa-garching.mpg.de/1137699/hl202602`

[9] F. W. Hehl *et al.*, "General relativity with spin and torsion," Rev. Mod. Phys. **48**, 393 (1976). doi: 10.1103/RevModPhys.48.393

[10] S. Weinberg, *The cosmological constant problem*, Rev. Mod. Phys. **61**, 1 (1989). doi:10.1103/RevModPhys.61.1

[11] J. Martin, *Everything you always wanted to know about the cosmological constant problem (but were afraid to ask)*, C. R. Phys. **13**, 566 (2012). doi:10.1016/j.crhy.2012.04.008, arXiv:1205.3169.